ENERGY CONSUMPTION

DATA ANALYTICS PROJECT

A Comprehensive Analysis of India's Energy Consumption (2019–2020)

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1. Executive Summary

This report documents a comprehensive data analytics project focused on analyzing energy consumption patterns across India for the years 2019 and 2020. The project covers the full data analytics pipeline — from raw data ingestion and SQL-based cleaning to interactive visualization using Tableau and web-based dashboard integration via Flask.

The primary objective was to uncover state-wise and region-wise energy usage trends, understand the impact of the COVID-19 lockdown in 2020, and provide interactive dashboards to facilitate data-driven decision-making. The project successfully delivered 14 Tableau visualizations, 3 interactive dashboards, and a public-facing web application.

2. Project Overview

2.1 Background

Energy consumption data is critical for infrastructure planning, policy decisions, and sustainability efforts. India, with its diverse geographical regions and states, presents a rich dataset for analyzing how electricity usage varies by region, state, season, and time period. The year 2020 introduced a unique variable — the COVID-19 nationwide lockdown — which significantly altered consumption patterns.

2.2 Objectives

- Analyze state-wise and region-wise energy consumption across India
- Compare 2019 and 2020 energy usage to measure the COVID-19 lockdown impact
- Identify top and bottom consuming states
- Visualize monthly, quarterly, and yearly consumption trends
- Create interactive dashboards with drill-down navigation
- Deploy the Tableau dashboard within a Flask web application

2.3 Tools & Technologies

Tool / Technology	Purpose
MySQL Workbench	Data ingestion, cleaning, and transformation using SQL
Tableau Desktop / Public	Interactive data visualization and dashboard creation
Python (Flask)	Web framework to embed and serve the Tableau dashboard

Tool / Technology	Purpose
GitHub	Version control and project repository management
CSV (Consumption.csv)	Source dataset containing raw energy consumption data

3. Dataset Description

3.1 Source Files

- Consumption.csv — Original raw dataset
- Refined_dataset.csv — Cleaned and formatted dataset post SQL processing

3.2 Dataset Columns

Column Name	Description
states	Name of the Indian state
regions	Geographical region (North, South, East, West, etc.)
latitude	Latitude coordinate of the state
longitude	Longitude coordinate of the state
dates	Date of the energy consumption record
usage	Recorded energy consumption value (in units)

3.3 Data Scope

The dataset covers energy consumption records across all major Indian states, categorized by region. It spans the calendar years 2019 and 2020, enabling year-over-year comparison and month-wise trend analysis. Geographic coordinates (latitude/longitude) enabled map-based visualizations in Tableau.

4. Data Preparation & Cleaning

4.1 Importing Data into MySQL

The raw CSV file (Consumption.csv) was imported into MySQL Workbench as the first step in the data pipeline. This established a structured database environment for applying SQL transformations.

4.2 SQL Cleaning Steps

The following data cleaning operations were performed using SQL queries (documented in data_cleaning.sql):

1. Imported CSV dataset into MySQL Workbench
2. Cleaned dataset using custom SQL queries
3. Checked for and handled NULL/missing values
4. Filtered relevant records required for visualization
5. Reformatted the dates column to ensure consistent date formatting
6. Verified the refined data was ready for Tableau integration

4.3 Output

The refined output was saved as Refined_dataset.csv, with the dates column properly formatted and all data quality issues resolved. This clean dataset was then connected to Tableau for visualization.

5. Tableau Visualization

5.1 Dataset Connection

After cleaning, the Refined_dataset.csv was connected to Tableau. The connection was established through MySQL, and all six fields were confirmed and verified within Tableau's data source interface: state, region, latitude, longitude, dates, and usage.

5.2 Visualizations Created

A total of 14 individual visualizations were created, as detailed below:

#	Visualization Name	Purpose
1	2019 Energy Consumption	Annual consumption overview for 2019
2	2020 Energy Consumption	Annual consumption overview for 2020

#	Visualization Name	Purpose
3	Total Consumption	Combined 2019 + 2020 total energy usage
4	Usage by Region	Regional breakdown of energy consumption
5	Top N States	Highest consuming states (filter-enabled)
6	Bottom N States	Lowest consuming states (filter-enabled)
7	2019 Month-wise Consumption	Monthly trend analysis for 2019
8	2020 Month-wise Consumption	Monthly trend analysis for 2020
9	Usage Before and After Lockdown	COVID-19 lockdown impact analysis
10	Total Regional Consumption (Pie Chart)	Proportional regional share of consumption
11	Region-wise Usage	Detailed regional comparison
12	State Usage	State-level usage detail
13	Quarter Usage	Quarterly consumption trends
14	Usage by Year	Year-over-year comparison (2019 vs 2020)

5.3 Filters & Enhancements

- Filters were applied to multiple visualizations for improved accuracy and user control
- Top N / Bottom N state filters allow dynamic ranking exploration
- Date filters enable granular month-wise and quarter-wise analysis

5.4 Dashboards

Three interactive dashboards were created, each consolidating related visualizations:

- Dashboard 1 — National overview: total and regional consumption
- Dashboard 2 — State-level analysis: top/bottom states, state usage comparisons
- Dashboard 3 — Temporal analysis: monthly, quarterly, and lockdown impact views

Each dashboard was integrated with navigation controls to allow seamless movement between dashboards. This ensured a guided, story-like user experience.

5.5 Story

A Tableau Story was created to present the visualizations in a narrative sequence. The story follows this arc: Energy Consumption Across India in 2019 and 2020 — beginning with a national overview, drilling into regional and state-level patterns, and concluding with the measurable impact of the COVID-19 lockdown on energy usage.

5.6 Public Dashboard

The completed Tableau workbook was published to Tableau Public and is accessible at the following URL:

https://public.tableau.com/views/ENERGYCONSUMPTIONVISUALIZATION/Dashboard1

6. Web Integration (Flask Application)

6.1 Overview

To extend the accessibility of the Tableau dashboard, the project included a web integration phase using Python and Flask. This allowed the Tableau Public dashboard to be embedded inside a fully functional web application.

6.2 Technologies Used

- Python — Core backend programming language
- Flask — Lightweight web framework for serving HTML pages
- Tableau Public Embed URL — Used to render the dashboard as an iframe within the app

6.3 Application Structure

The Flask web application (97% HTML, 3% Python by language composition) served an embedded Tableau dashboard view. The embed approach used Tableau Public's share link to render the interactive dashboard directly in the browser without requiring a Tableau license from end users.

7. Key Findings & Insights

7.1 Regional Consumption Patterns

Analysis of the dataset revealed significant variation in energy consumption across India's regions. The pie chart and regional breakdown visualizations showed that certain high-

industrialization regions consistently outpaced others in total consumption, reflecting the concentration of industrial and commercial activity.

7.2 COVID-19 Lockdown Impact (2020)

The 'Usage Before and After Lockdown' visualization provided clear evidence of consumption reduction during the March–May 2020 lockdown period. Industrial activity halted and commercial consumption dropped sharply, while residential consumption showed a modest uptick — consistent with global patterns observed during similar lockdowns.

7.3 Monthly & Seasonal Trends

Month-wise visualizations for both 2019 and 2020 revealed seasonal consumption peaks, particularly in summer months (March–June) driven by cooling demand, and relatively lower consumption in post-monsoon months. This pattern was significantly disrupted in 2020 due to the lockdown.

7.4 Top & Bottom Consuming States

The Top N and Bottom N state analysis identified the highest and lowest energy consumers among Indian states. Highly industrialized and populous states predictably appeared in the top tier, while smaller or predominantly agrarian states appeared at the lower end of the consumption spectrum.

8. Project Repository Structure

File / Folder	Type	Description
Consumption.csv	Data	Raw energy consumption dataset
Refined_dataset.csv	Data	Cleaned and formatted dataset
data_cleaning.sql	SQL Script	SQL queries for data preparation
ENERGY CONSUMPTION VISUALIZATION.twbx	Tableau	Packaged Tableau workbook with all visualizations
DATA ANALYTICS PROJECT/	Folder	Core project files
DATA VISUALIZATION USING TABLEAU SCREENSHOTS/	Folder	Screenshots of all Tableau visualizations
Screenshots/	Folder	Additional project screenshots

File / Folder	Type	Description
README.md	Documentation	Project overview and setup instructions

9. Conclusion

This data analytics project successfully demonstrated a complete end-to-end analytics workflow: from raw data ingestion and SQL-based cleaning, through advanced Tableau visualization, to web-based deployment via Flask. The project delivers meaningful insights into India's energy consumption patterns, highlights the measurable impact of the 2020 COVID-19 lockdown, and presents the findings through intuitive, interactive dashboards.

The work reflects proficiency in data engineering (MySQL), business intelligence tooling (Tableau), and web development (Flask/Python), making it a well-rounded demonstration of practical data analytics skills.

9.1 Future Enhancements

- Extend the dataset to include post-2020 years for longer-term trend analysis
- Incorporate predictive modeling (e.g., time-series forecasting) for future consumption estimates
- Add geospatial map visualizations using Tableau's built-in mapping with latitude/longitude fields
- Build a full REST API backend to serve dynamic data queries to the Flask frontend
- Automate data refresh pipelines to keep the dashboard updated with real-time data